



TECHNICAL SERVICE BULLETIN

Engine Overheating And/Or Poor Air Conditioning (A/C) Performance With DTC P0480 And/Or P0217

26-2099

10 June 2026

This bulletin supersedes 24-2224. Reason for update: Add/Remove/Update Labor Operation/PDR, Revised Service Procedure (TSB/GSB) / Text (SSM)

Model:

Ford 2022-2024 Everest	Build Date: Up to 18 March 2024 Assembly Plant: AAT- RAYONG ASSEMBLY PLT-LS (AAGDZ)
2022-2024 Ranger	Build Date: Up to 18 March 2024 Assembly Plant: AAT- RAYONG ASSEMBLY PLT-LS (AAGDZ)
2022-2024 Ranger	Build Date: Up to 18 March 2024 Assembly Plant: THAILAND LTD PLANT BUILD (AAGB7)
2022-2024 Ranger	Build Date: Up to 16 April 2024 Assembly Plant: HAIDUONG ASSEMBLY PLT(VIETNAM) (AAGBS)
2022-2024 Ranger	Assembly Plant: SILVERTON(S.AFRICA)PLANT BLD (AAGBQ)

Issue: Some 2022-2024 Next-Generation Ranger, Ranger Raptor and Everest vehicles may experience engine overheating and/or poor A/C performance with Diagnostic Trouble Code (DTC) P0480 and/or P0217. This may be caused by an inoperative cooling fan due to damage to the cooling fan relay and wiring.

Action: Follow the Service Procedure below on vehicles that meet all of the following criteria:

- One of the following vehicles:
 - 2022–2024 Next-Generation Ranger AAT (Thailand), built to 18-Mar-2024
 - 2022–2024 Next-Generation Everest AAT (Thailand), built to 18-Mar-2024
 - 2022–2024 Next-Generation Ranger FTM (Thailand), built to 18-Mar-2024
 - 2022–2024 Next-Generation Ranger HAIDUONG ASSEMBLY PLT(VIETNAM), built to 16-Apr-2024
 - 2022–2024 Next-Generation Ranger SILVERTON (S.AFRICA) PLANT BLD
- Engine overheating and/or poor A/C performance with DTC P0480 and/or P0217

Parts

Service Part Number	Claim Quantity	Package Order Quantity	Number in Package	Description
MB3Z-14S411-E	1	1	1	Kit Wiring

Labor Times

Description	Operation No.	Time
Connect the Ford Diagnosis and Repair System (FDRS) to the vehicle and check for DTC (includes - radiator sight shield removal and relay, relay terminal inspection)	262099A	0.4 Hrs.
Replace/Install both the cooling fan relay and terminal connector using the released service pigtail kit (repair wiring as per the given service procedure and confirm cooling fan operation)	262099B	1.0 Hrs.
Check cooling fan operation and check and clear any DTCs stored in the system	262099C	0.1 Hrs.

Repair/Claim Coding

Causal Part:	14290
Condition Code:	46

PDR Table

PDR Code	Description	Time
TSB262099	Connect the Ford Diagnosis and Repair System (FDRS) to the vehicle and check for DTC and Replace/Install both the cooling fan relay and terminal connector using the released service pigtail kit and Check cooling fan operation and check and clear any DTCs stored in the system	1.5 Hrs.

Diagnosis Procedure

1. Connect the Ford Diagnosis and Repair System (FDRS) to the vehicle and check for DTC P0480 and/or P0217.
2. Remove the bolt and the air cleaner intake tube. (Figure 1)

Torque: 10.5 Nm.

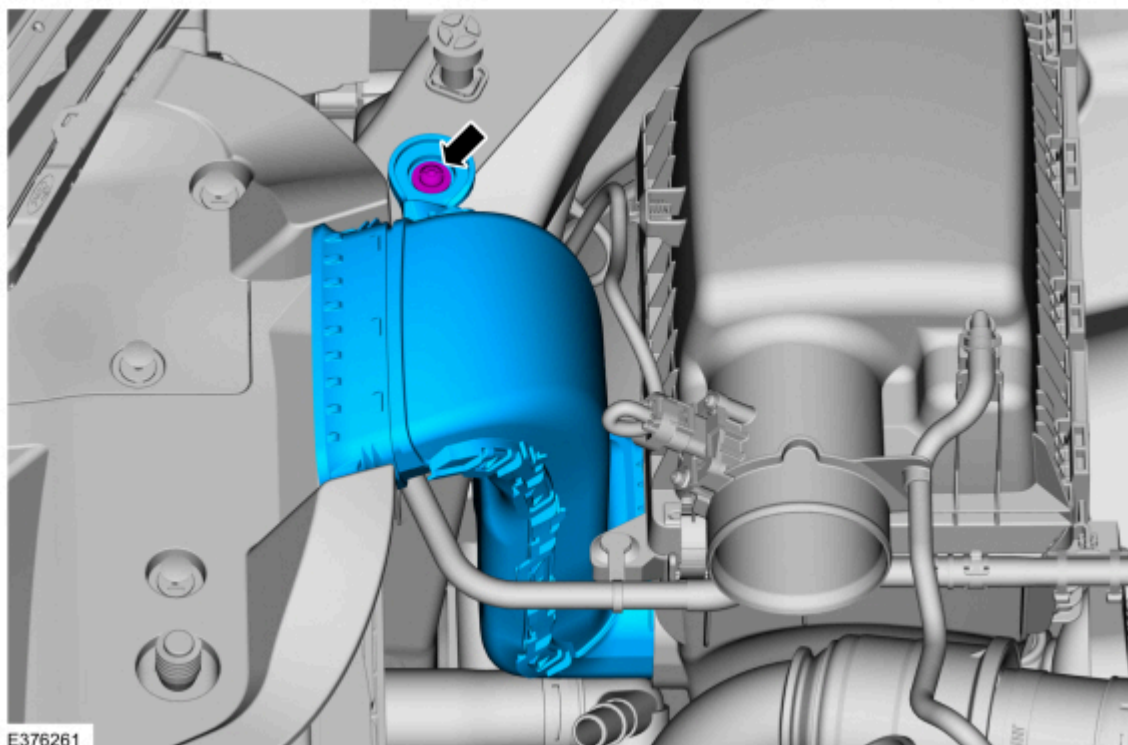


Figure 1

3. Remove the push pins and the radiator sight shield. (Figure 2)

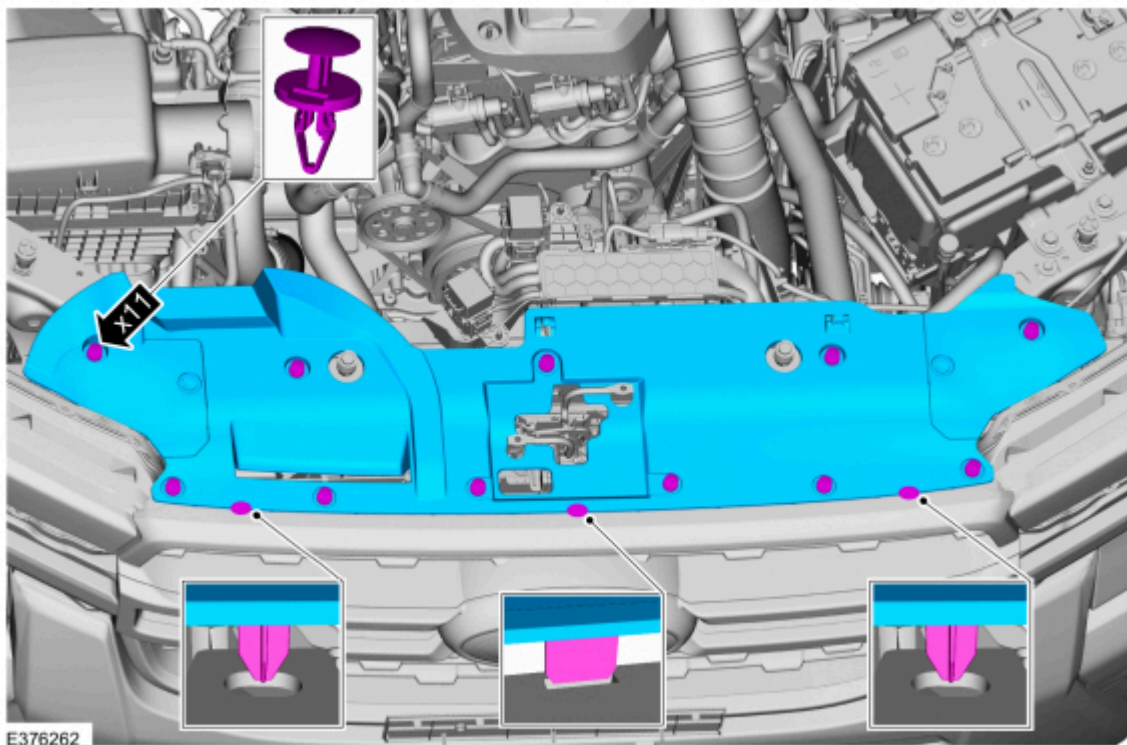


Figure 2

4. Inspect the cooling fan relay part number.

- (1). If the installed relay is part number F80B-14B192-AA (Figure 3) – go to Step 5.
- (2). If the installed relay is part number RU5T-14B192-FEA (Figure 4) - this article does not apply. Refer to the Workshop Manual (WSM) DTC or Symptom Chart for diagnosis.



E440894

Figure 3 – Tin Relay F80B-14B192-AA



E440895

Figure 4 – Silver Relay RU5T-14B192-FEA

5. Is DTC P0480 and/or P0217 logged or is there any visible damage to the cooling fan relay and/or relay holder?

(1). Yes - proceed to Step 6.

(2). No - this article does not apply. Refit the radiator sight shield and air cleaner intake tube. Refer to the WSM DTC or Symptom Chart for diagnosis.

6. Refer to WSM Section 303-03 Engine Cooling, Cooling Fan Control, Pinpoint Test KN for these DTCs, paying special attention to the condition of the cooling fan relay and relay holder.

NOTE: The illustrations below are a guide only. In some cases, the damage to the relay and/or relay holder may not be as immediately obvious as that shown in the illustrations, careful inspection is recommended. (Figure 5 to 7)

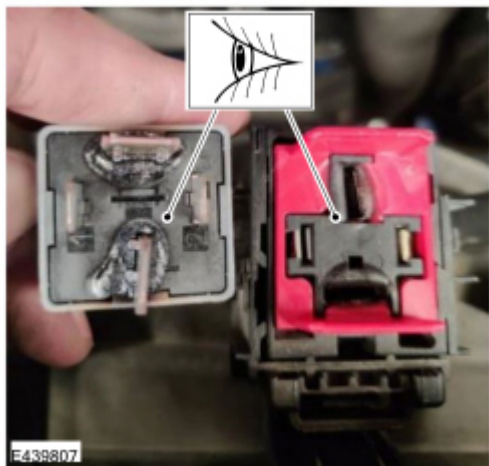


Figure 5 - Cooling fan relay and wiring damage

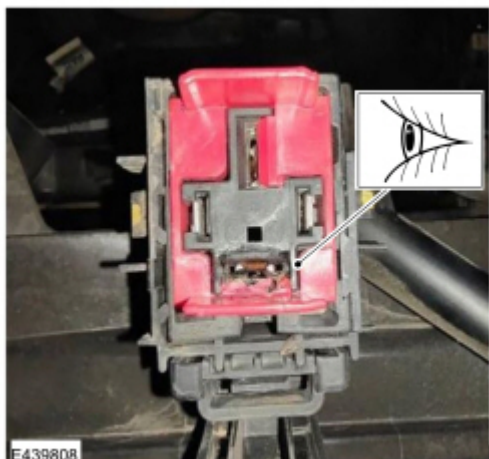


Figure 6 - Cooling fan relay and wiring damage - Connector side



Figure 7 - Cooling fan relay and wiring damage - Relay side

7. Is any damage to the cooling fan relay or terminal connector identified, similar to that shown in Figure 5 to 7 above?

(1). Yes - proceed to Step 8.

(2). No - this article does not apply. Refit the radiator sight shield and air cleaner intake tube. Refer to the WSM DTC or Symptom Chart for diagnosis.

8. Replace both the cooling fan relay and terminal connector using the released service pigtail kit Part Number MB3Z-14S411-E (Figure 8). Refer to the **Wiring Repair and Splice Operation Procedure** below and the recommended splicing method shown in the Professional Technician System (PTS) - Wiring tab - Cell 005 Connector Repair Procedures - Pages 5 to 9 and refer to the Wiring Pigtail Kit/Wiring Terminal Kit Guide shown in PTS.

Callout 1: Cable Ties

Callout 2: Heat Shrink Seal Tubes

Callout 3: Small Wiring Crimp Qty 4

Callout 4: Large Wiring Crimp Qty 4

Callout 5: New Relay Part Number: RU5T-14B192-FEA

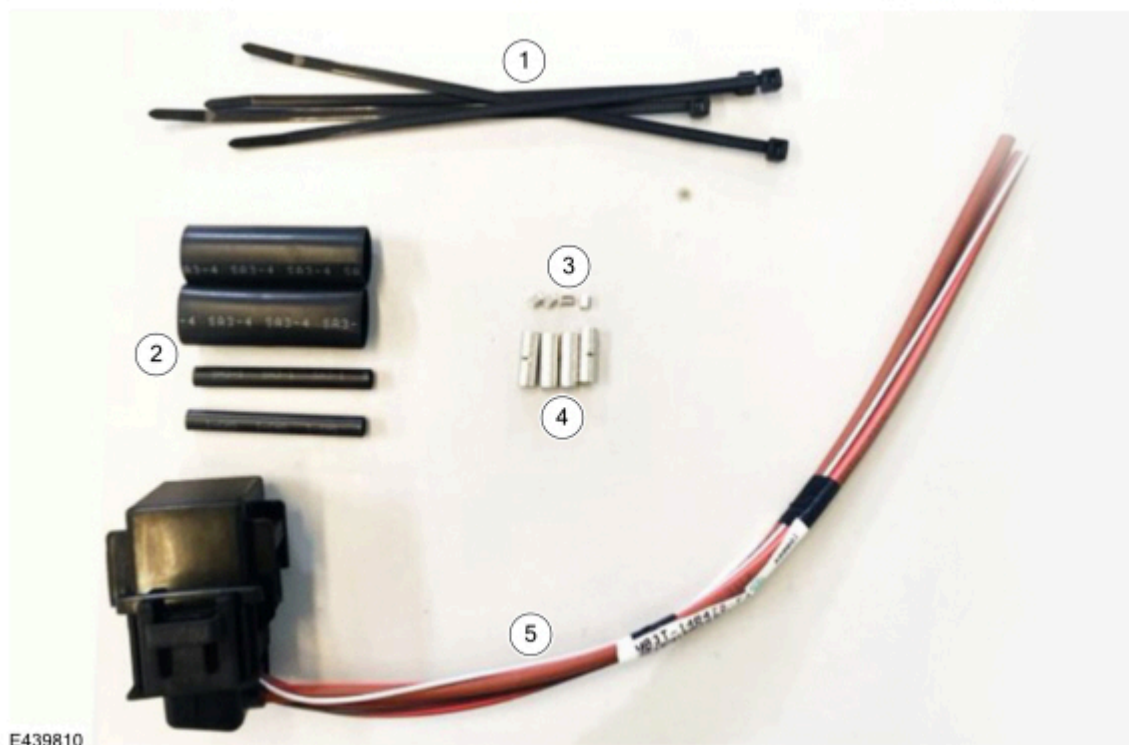


Figure 8

NOTE: When completing a repair under this TSB it is essential to ensure that **BOTH** the relay and the pigtail connector are replaced with the improved parts provided in the service kit. **BOTH** the relay and the pigtail connector **MUST** be replaced to ensure a robust repair on the vehicle.

Wiring Repair and Splice Operation Procedure:

1. Disconnect the battery cable. Refer to WSM Section 414-01.
2. Access the cooling fan relay location. (Figure 9)

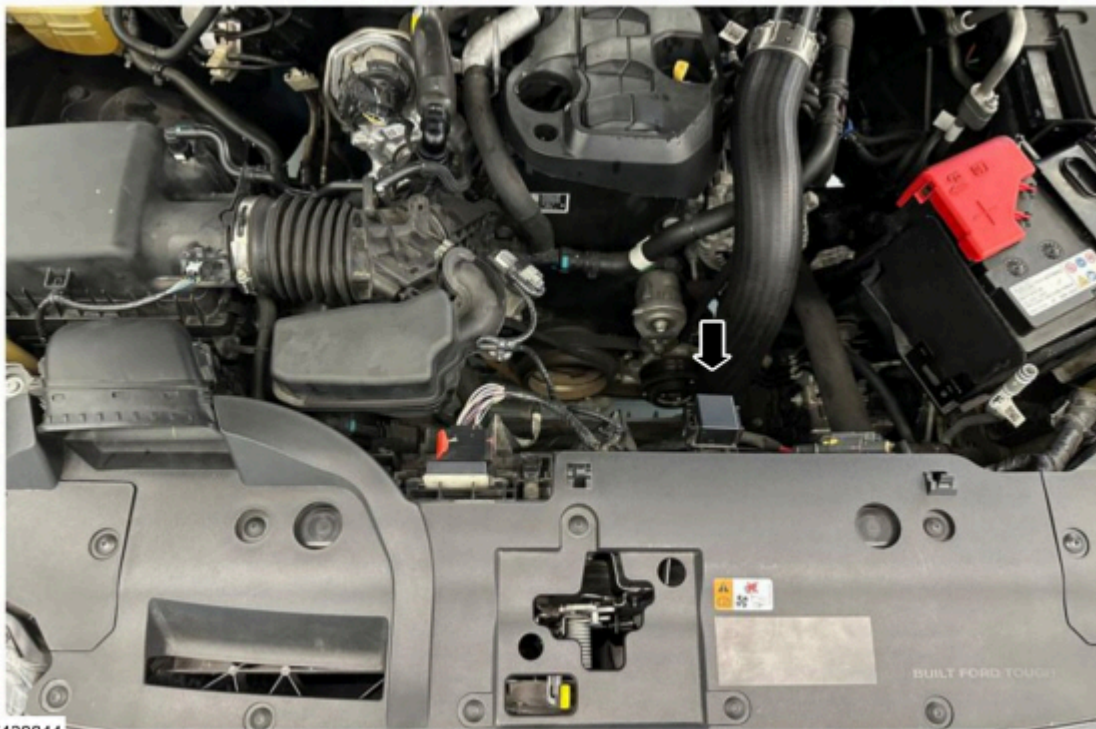


Figure 9

3. Cut the cable tie to remove the cooling fan connector for repair. (Figure 10)

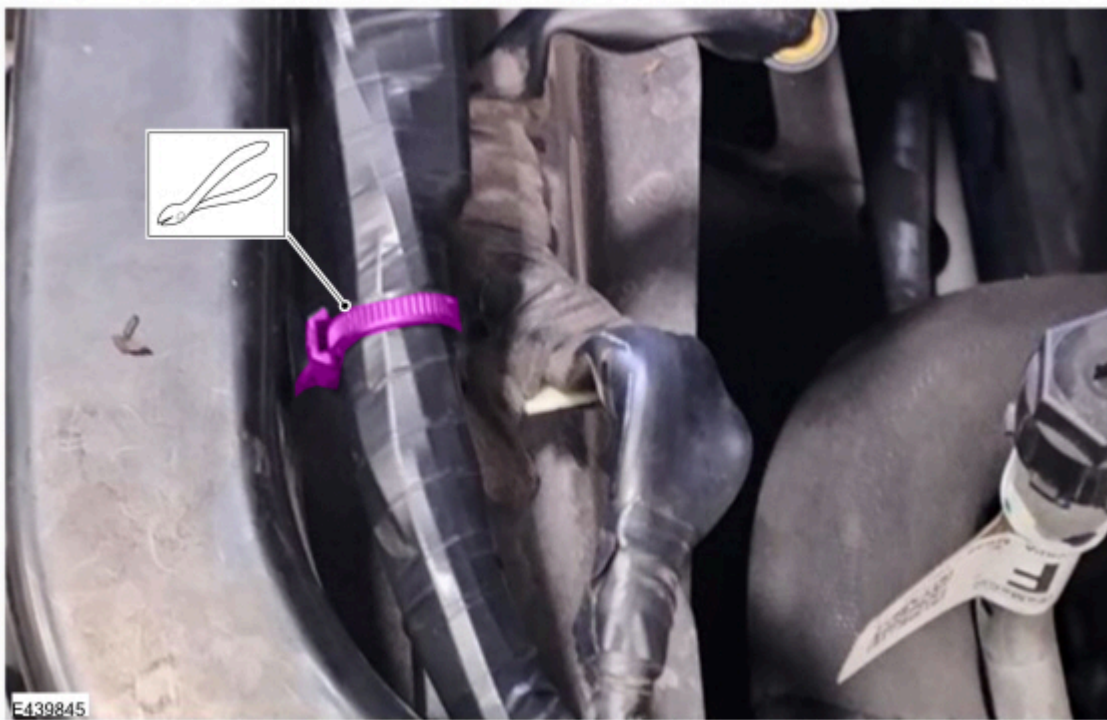


Figure 10

NOTE: Some vehicles have connector C1026 (connector for the Anti-Lock Brake System (ABS) attached on the cooling fan wiring. Technicians are advised to measure and record the distance between the C1026 connector to the wiring grommet to ensure installing the cooling fan wiring in the same position after the repair. (Figure 11)

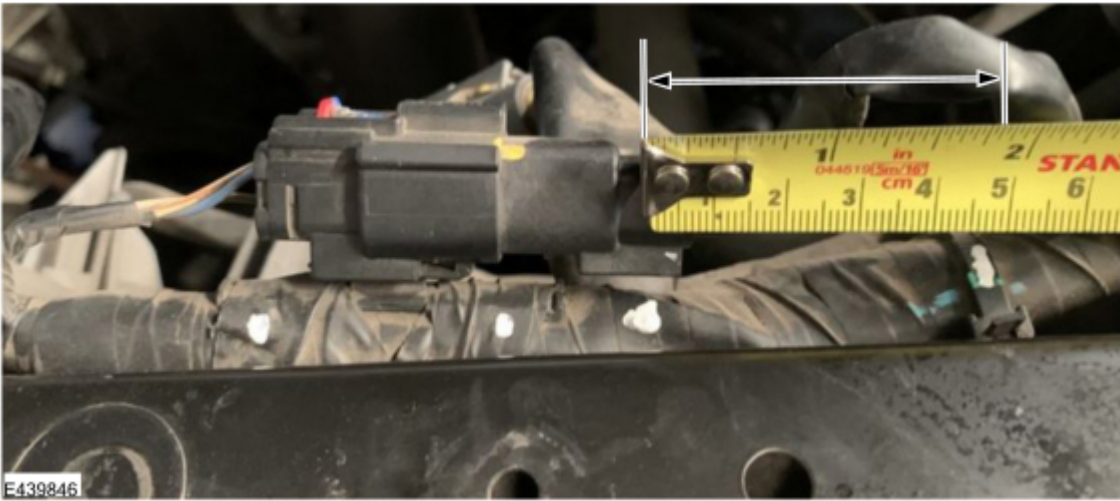


Figure 11

4. Remove the cooling fan relay. (Figure 12)

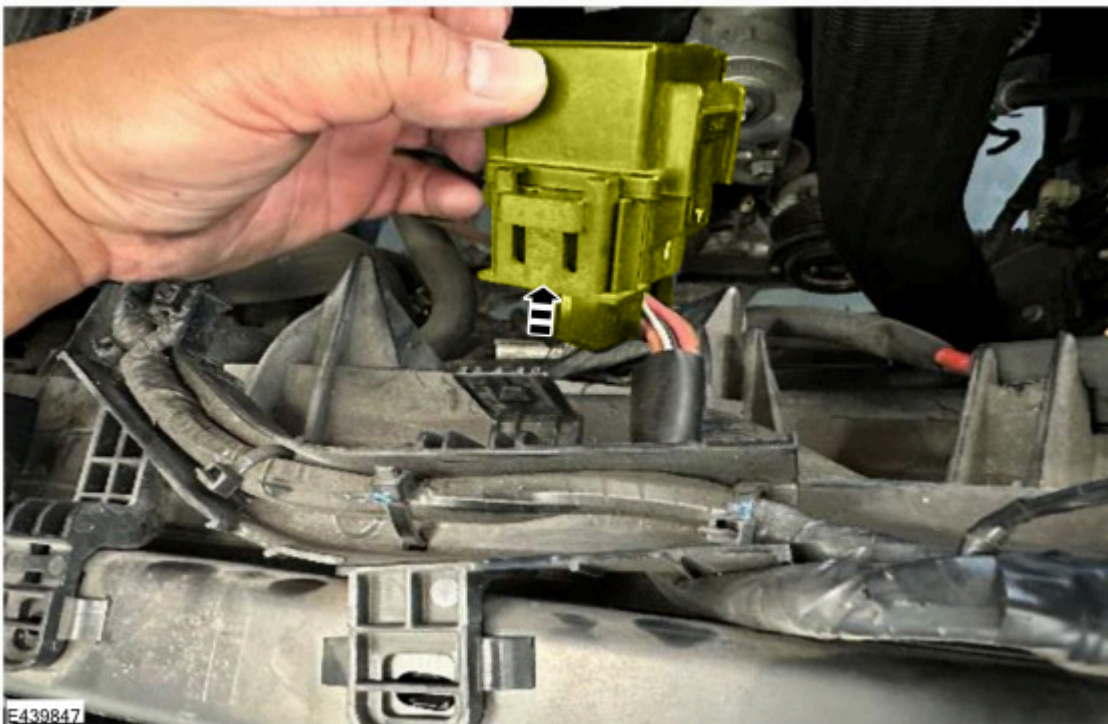


Figure 12

5. Remove the PVC tube and tape on the cooling fan wiring to repair (only the affected area). (Figure 13)

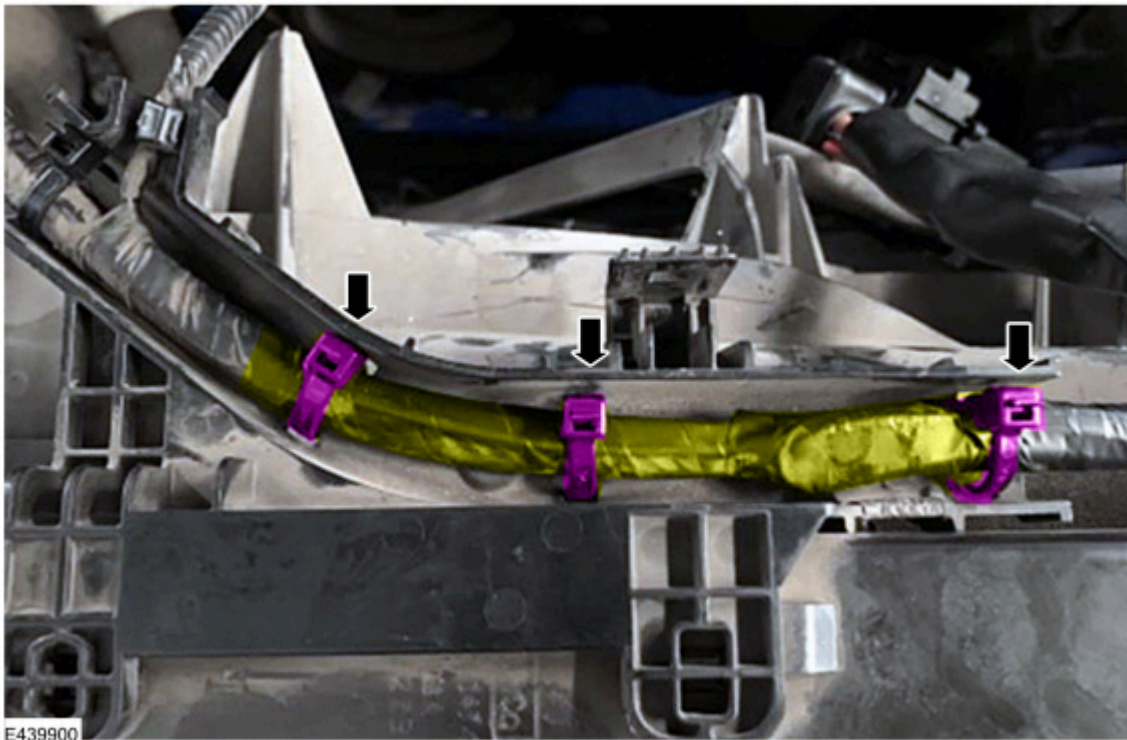


Figure 13

6. Remove the PVC tape on the wiring to access the cooling fan wiring inside. (Figure 14)

NOTE: Peel the PVC tape only in the necessary area, do not go past the wiring grommet.

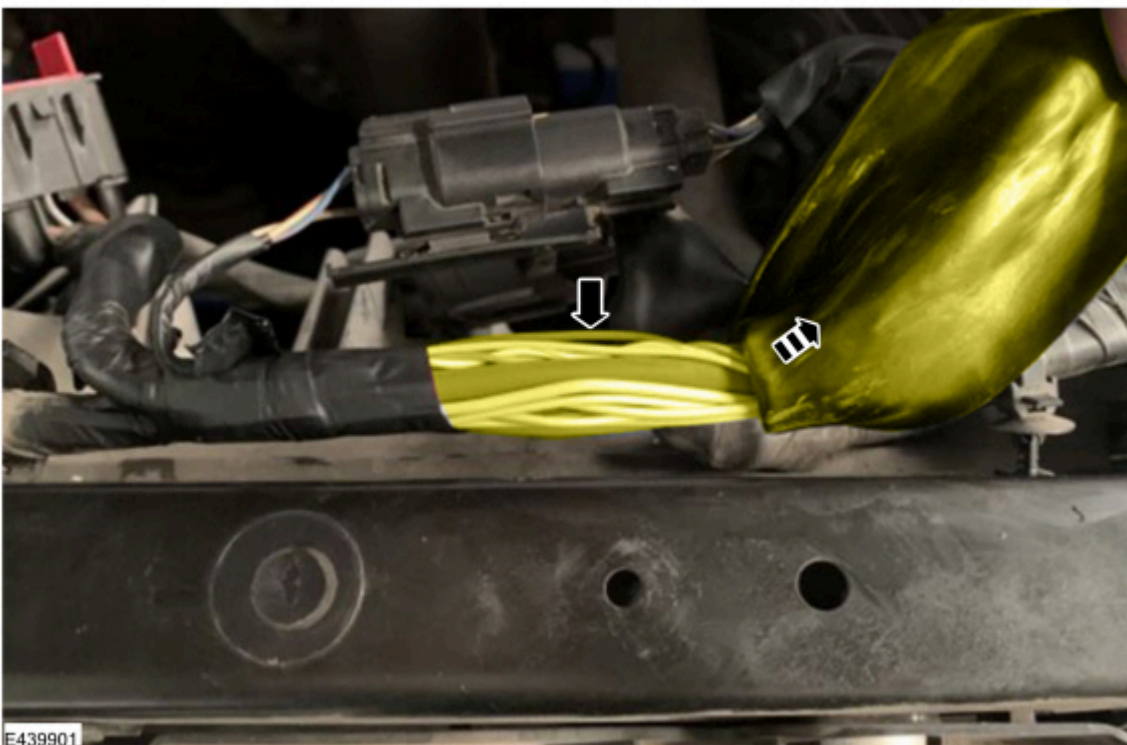


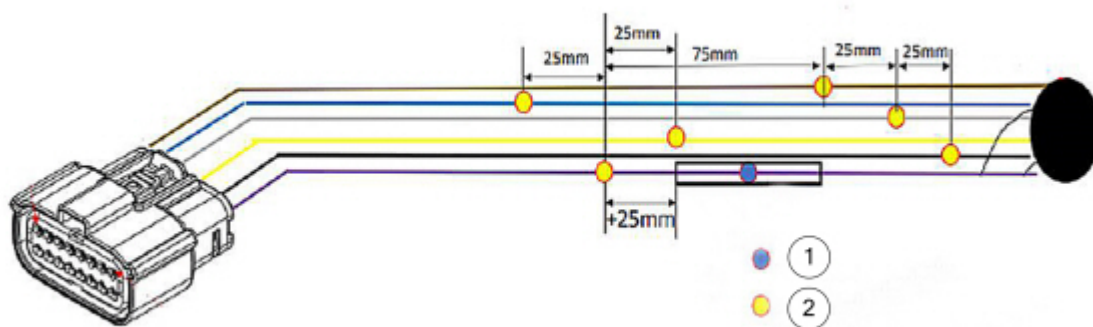
Figure 14

7. Mark and cut the 4 cooling fan wires as per Figure 16 below.

NOTE: The splices are done at different lengths to reduce wiring diameter. (Figure 15)

Callout 1: Exit Connection Joint

Callout 2: Pigtail Kit Connection Joint



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Figure 15

Callout 1: Wiring grommet

Callout 2: BLACK wire - Measure 70mm from the wiring grommet

Callout 3: 1st RED/BROWN wire - Measure 25mm from the BLACK wire cut position

Callout 4: 2nd RED/BROWN wire - Measure 25mm from the 1st RED/BROWN wire cut position

Callout 5: GRAY wire - Measure 25mm from the 2nd RED/BROWN wire cut position

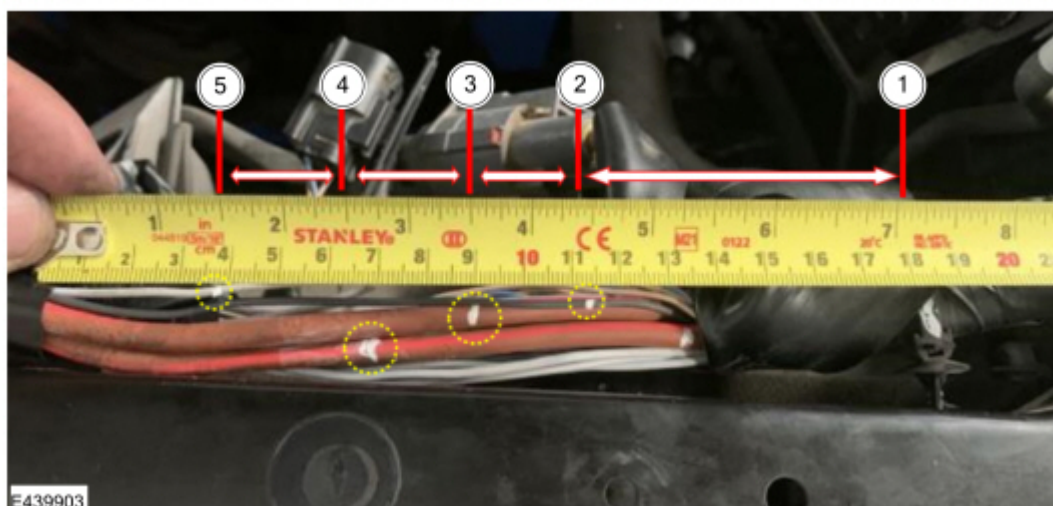


Figure 16

8. Mark the same cut positions on the pigtail kit using the cooling fan wiring as a reference. (Figure 17 and 18)

(1). Put the pigtail kit next to the cooling fan wiring to mark the distances.

(2). Cut the pigtail kit at the marked positions.

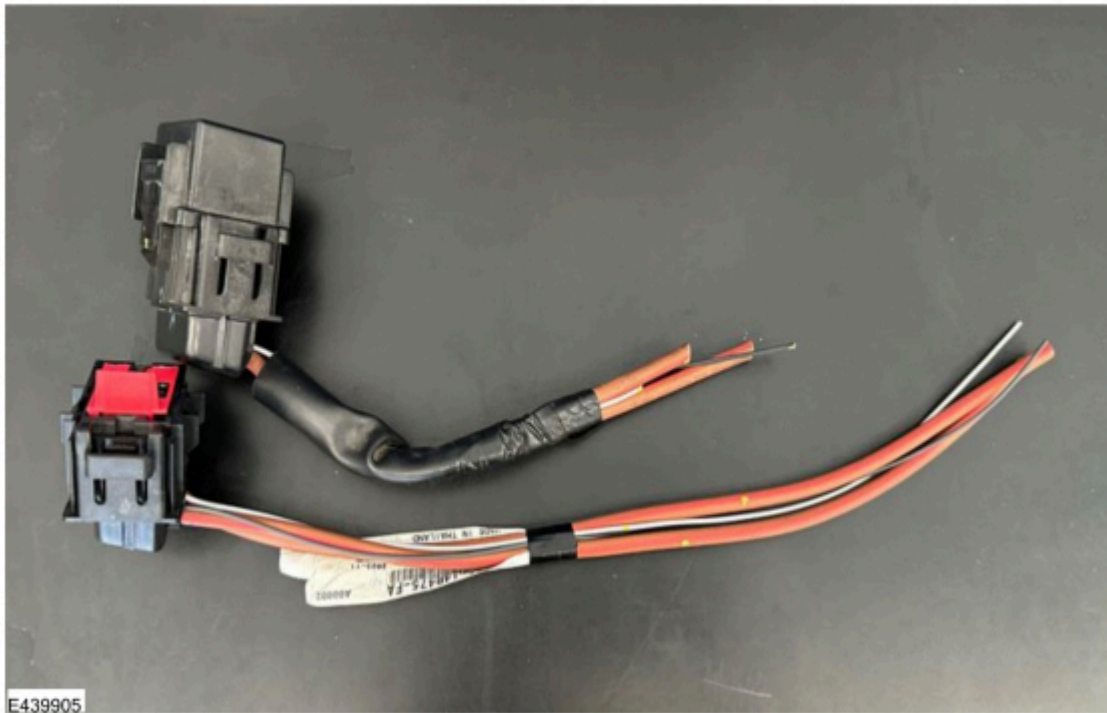


Figure 17



Figure 18

9. Repair the cooling fan wiring with the splicing pigtail kit following GSB 18AP/001 – Wiring Repair Job Aid, concentrating on repair method 4.2 - Crimping. The wires should be stripped of around 10mm of insulation. Suggested tools for the repair (other appropriate tools also ok to be used). (Figure 19)

Callout 1: Wire Stripper

Callout 2: Hand Crimp Tool

Callout 3: PVC Insulation Tape

Callout 4: Cutting Knife

Callout 5: Hot Air Gun



E439907

Figure 19

10. Place a heat shrink tube over each wire and use a hot air gun to seal the repair position (Figure 20). Inspect the heat shrink to ensure a seal with no gaps. (Figure 21)



E439908

Figure 20



Figure 21

11. Re-connect the battery cable. Refer to WSM Section 414-01.
12. Perform an electrical test and DTC check at this point BEFORE adding PVC and cable ties.
13. Seal the cooling fan wiring with PVC insulating tape and secure connector C1026 (where fitted) to the same position as identified in Step 3 above. (Figure 22 and 23)

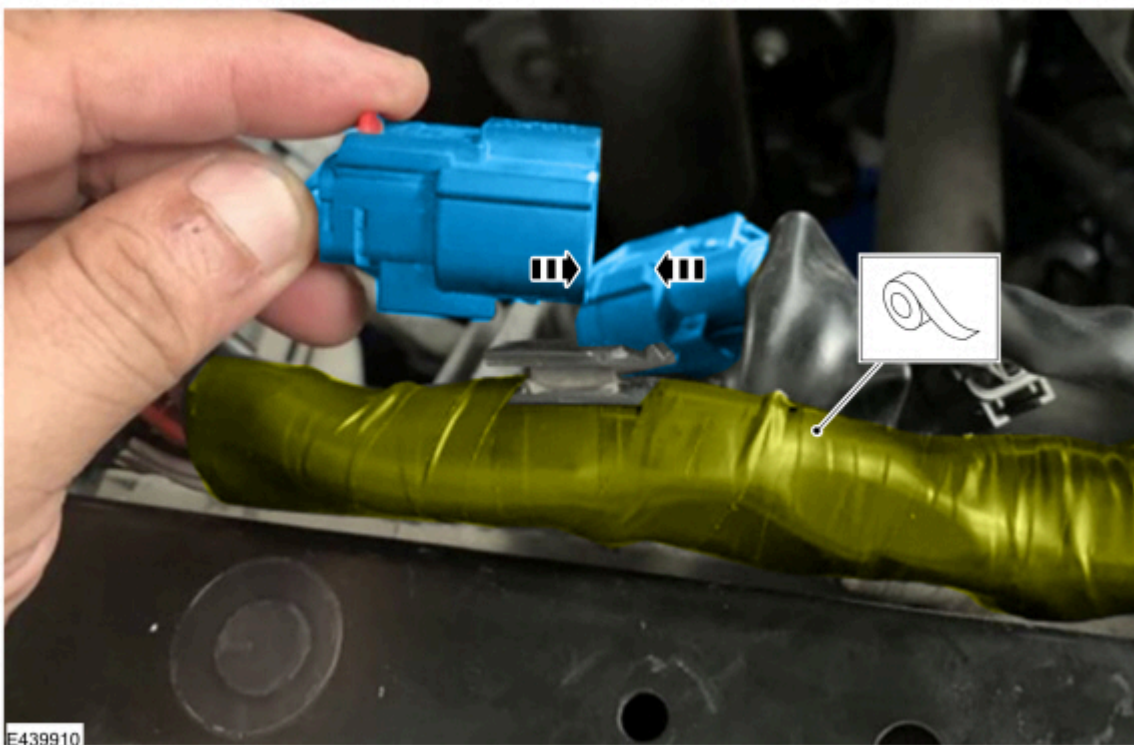


Figure 22



Figure 23

14. Re-install the wiring to the original position and ensure all connectors are connected correctly. (Figure 24 and 25)

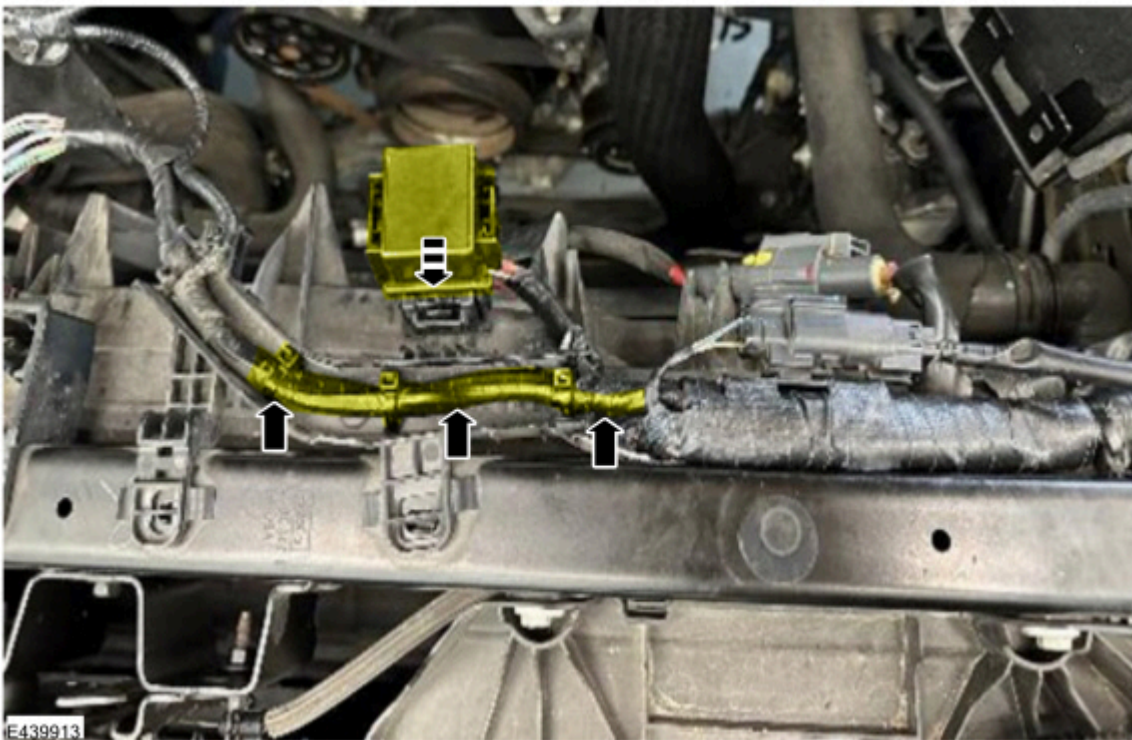


Figure 24



Figure 25

- 15.** Re-install the radiator sight shield and air cleaner intake tube. Reverse the removal procedure.
- 16.** Confirm cooling fan operation.
- 17.** Check and clear any DTCs upon completion of the repair.
- 18.** Return the vehicle to the customer.

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